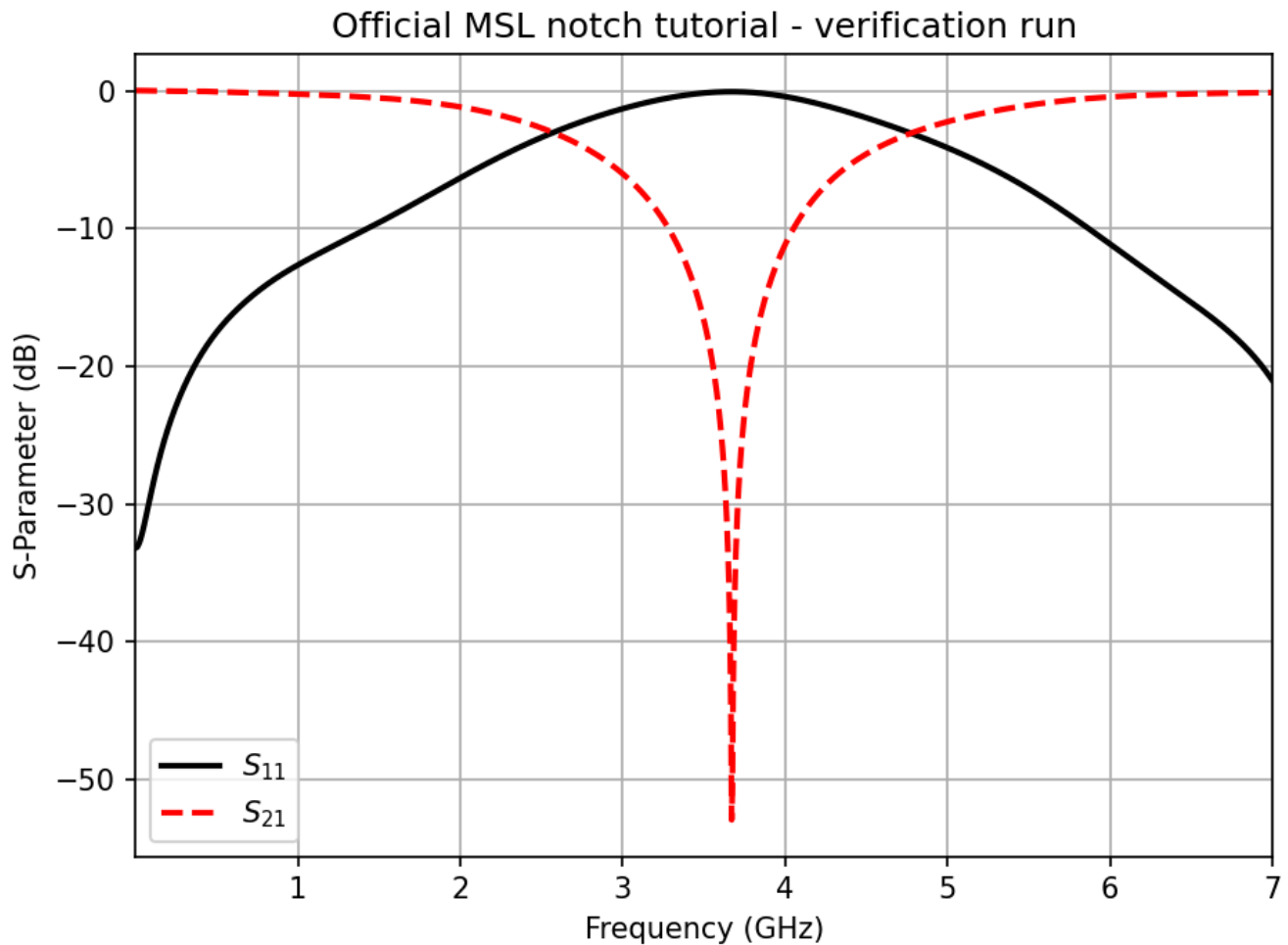


Methods in openEMS Validation

I initially started the openEMS sim by verifying the data file format, environment, and the MSL notch tutorial reproduction



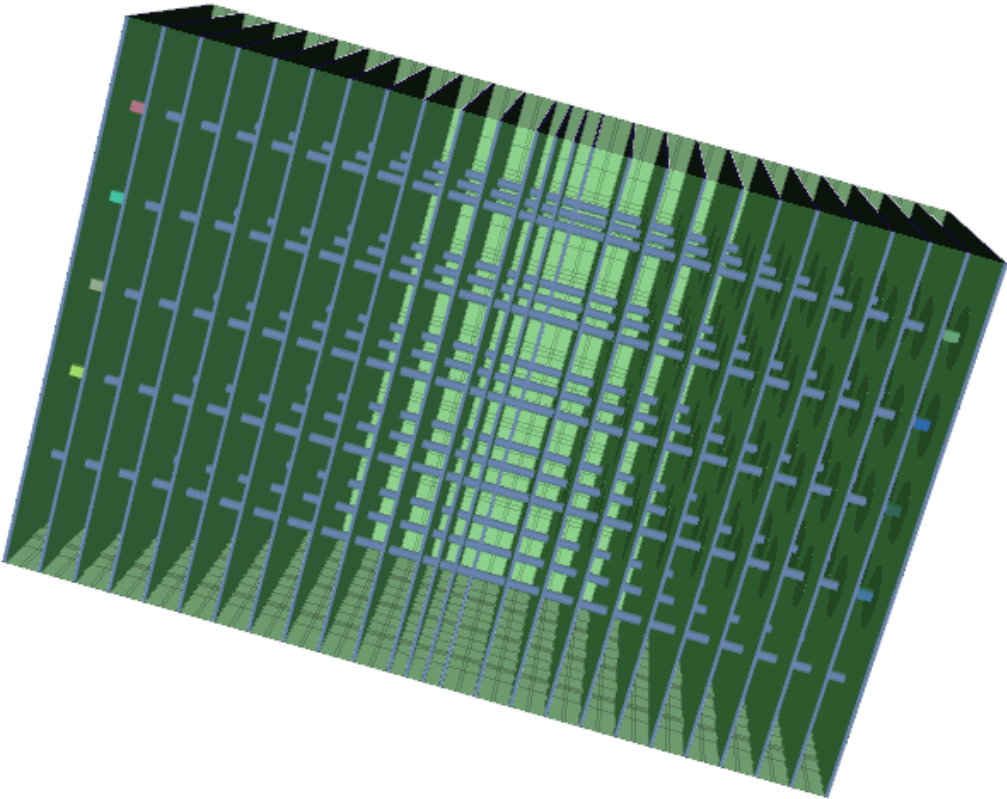
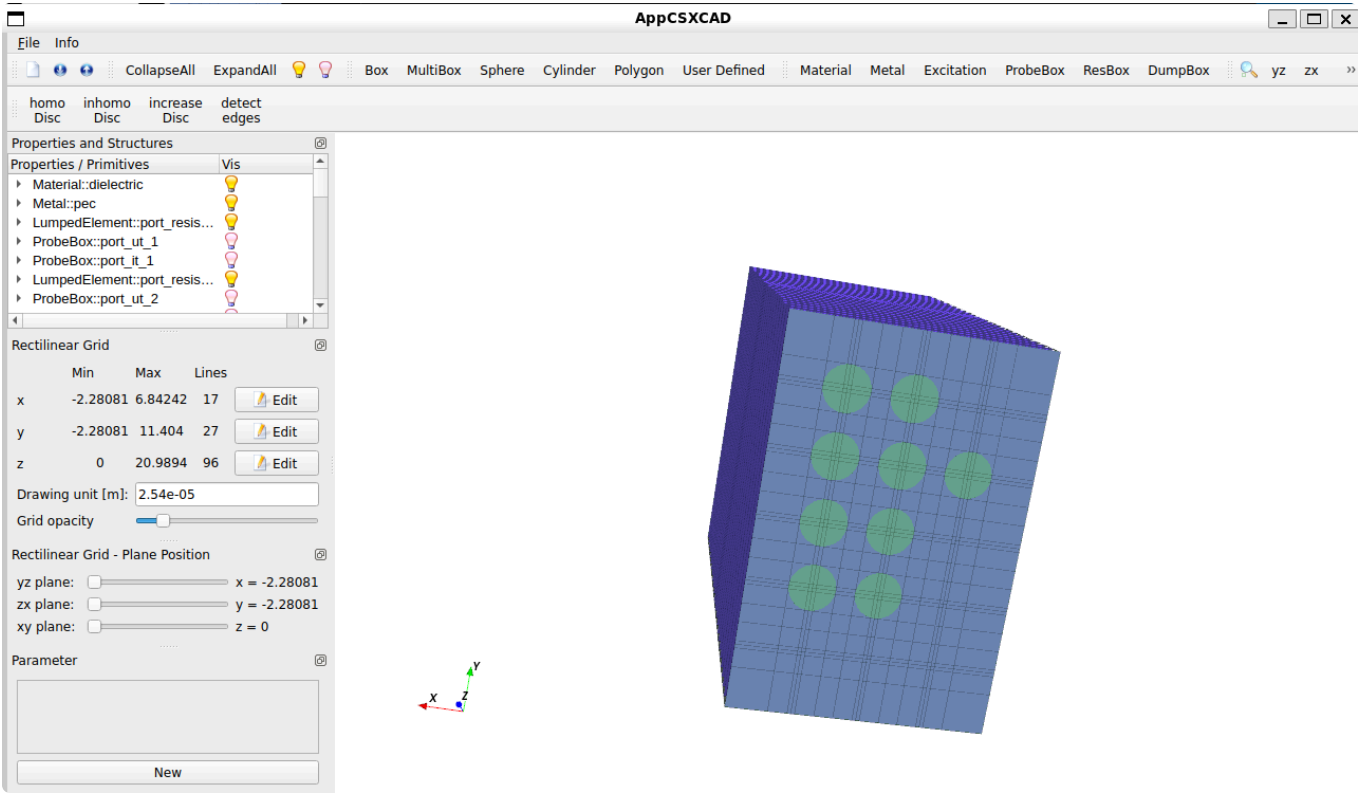
To keep out from geometry-truncation error and also match the CONMLS environment the entire via array was simulated and extract 4 port sub blocks. To make sure the data extraction from the processed differential pair data pipeline, I wrote a script to re-extract the differential pair 1 of the sim_okg_0017 (used for validation as it was the fastest simulation) directly from the raw .s16p file and then compare it with the processed data.

max |diff| over all 4x4xF : 3.574e-08

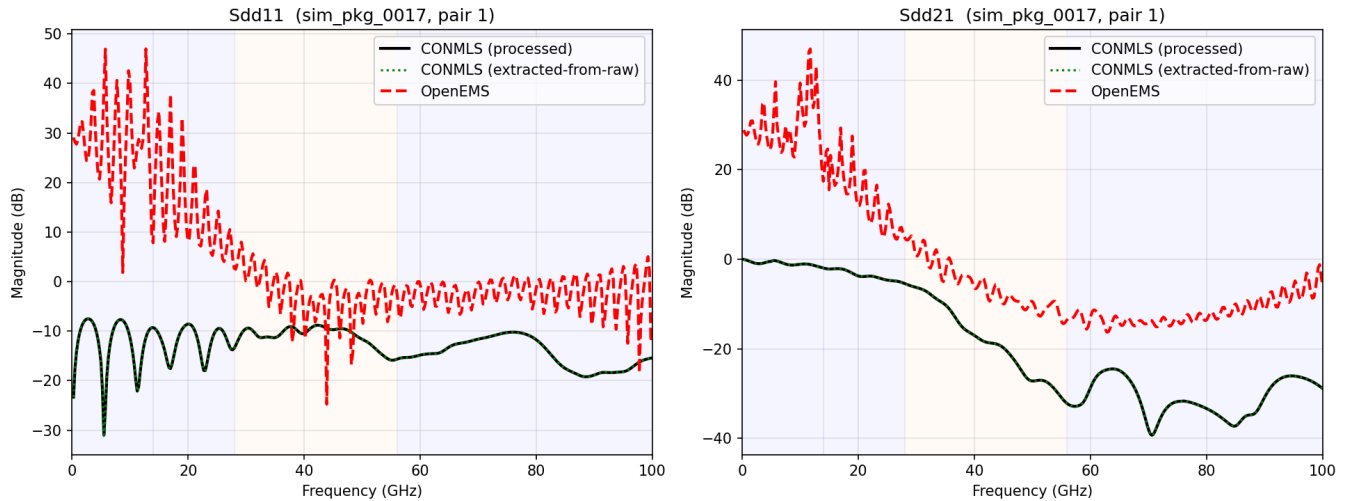
VERDICT: MATCH (extraction chain verified independent of solver)

there was a match with the processed dataset and the raw data - so I proceeded with the simulation script.

The openEMS 3d render on AppCSXCAD was looking good when I ran the first version of the script but it failed with the math.



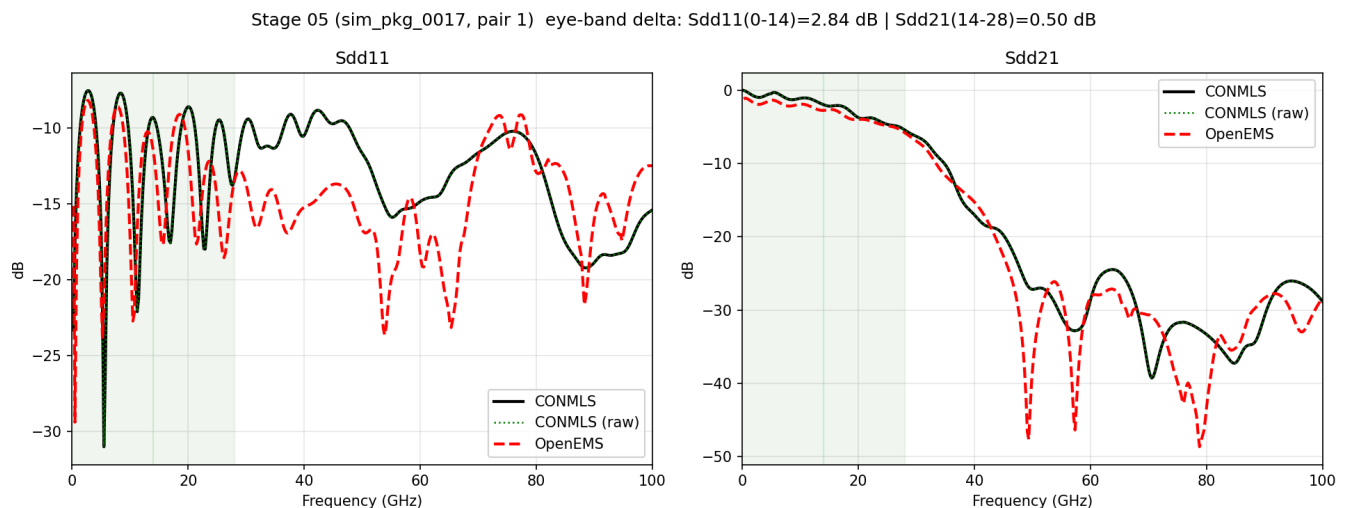
I initially tried to develop the script by using lumped ports centered on the via i.e. z-directed lumped port as a box of half-width r in x and y, centered on the via. But this failed. When I checked the logs, it was evident that at 0.25GHz, every one of the 16 ports reflected into itself. The port box was centred on the via and fully contained the PEC via cylinder.



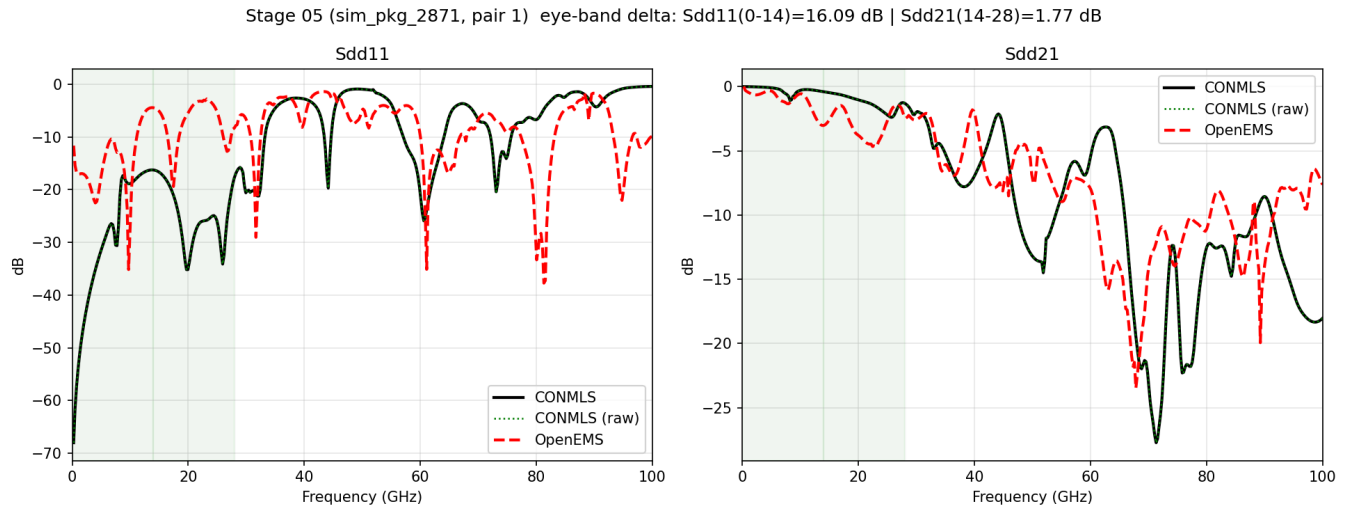
I later tried to troubleshoot this issue with different versions - trying out to fill the gap with a z-directed port and then radial ports across the antipad but they remained unstable with no major improvement to the simulation.

When I explored the documentation and other openEMS community channels (reddit), I got an idea of using CoaxialPort. the native Coaxial port was not available in the openEMS build version I was using so had to update it with the latest build and after trying out different variations in this - lossy coaxial stub fix, fixing the dielectric loss (KAPPA_ANCHOR_HZ = 28 GHz), by anchoring it to the Nyquist frequency by following the convention in the MSL tutorial, and replacing the PML_8 with MUR boundaries, the final version was fixed and was ready for validation.

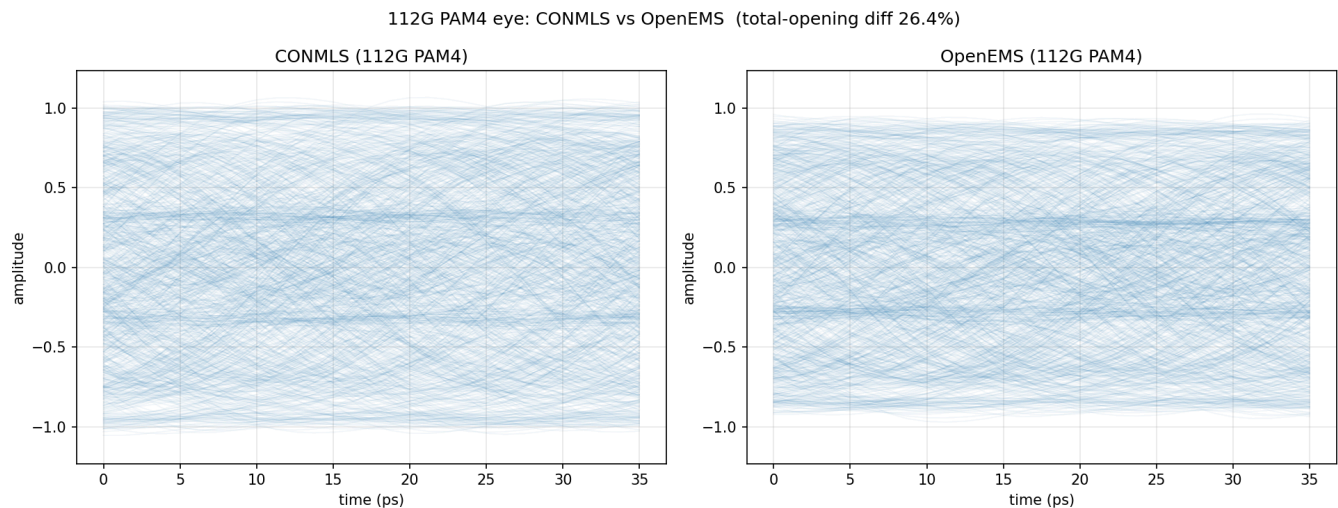
Tried `ref_plane_shift` improvement to check how s11 is improved



Even though PML_8 boundaries worked flawlessly for the 16 port sim (sim_pkg_0017) it exploded for the 20 port sim (sim_pkg_2871). So I changed the boundary to be MUR by default.



The eye diagram was then compared with the actual data (CONMLS) with the simulated data (openEMS)



eye for sim_pkg_0017

Later I wrote the script to compare the actual ground truth with the geometry generated by the Inverse model (baseline Geometry space TTO) and it was very evident that use of Geometry TTO led to surrogate exploitation because of non physicality and absence of physical constraint to maintain passivity. This was the motivation to explore more options on TTO as discussed on last meeting - To explore latent space TTO and Cumulative TTO with passivity filter.

Surrogate Exploitation clearly visible for the sameple sim_pkg_0017

Stage 07: design_sample_0 (template sim_pkg_0017, pair 1, geometry='x_local_norm')

--- geometry sanity check [design_sample_0] ---

via_radius = 2.403 mil

antipad_rad = 23.173 mil

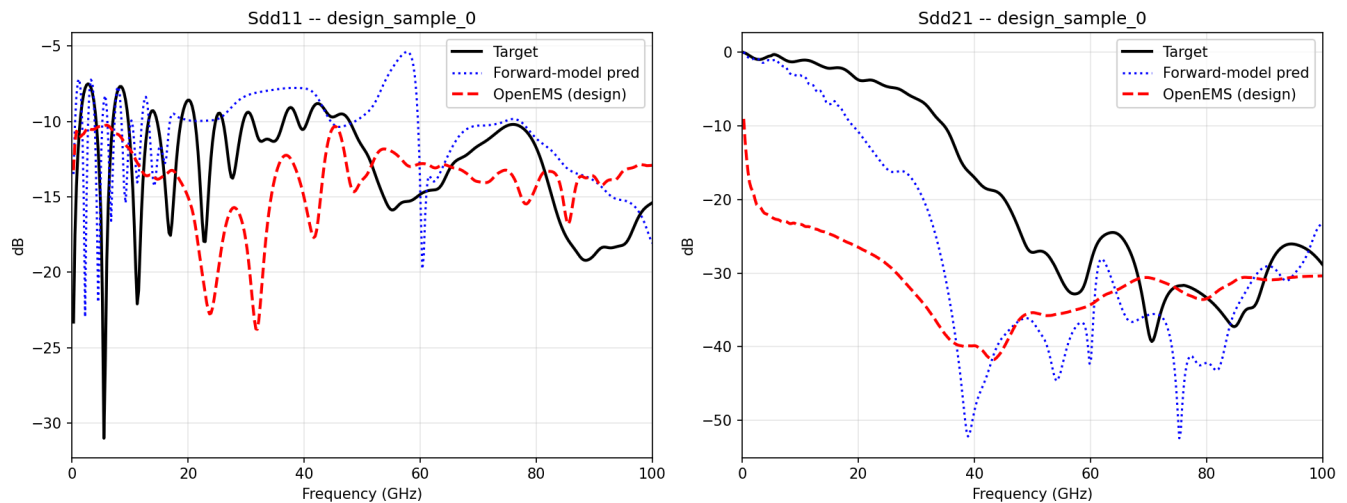
pitch = 66.312 mil

TDIEL/TMET = 1296.2 total, eps_r=3.383, tan_d=0.1039

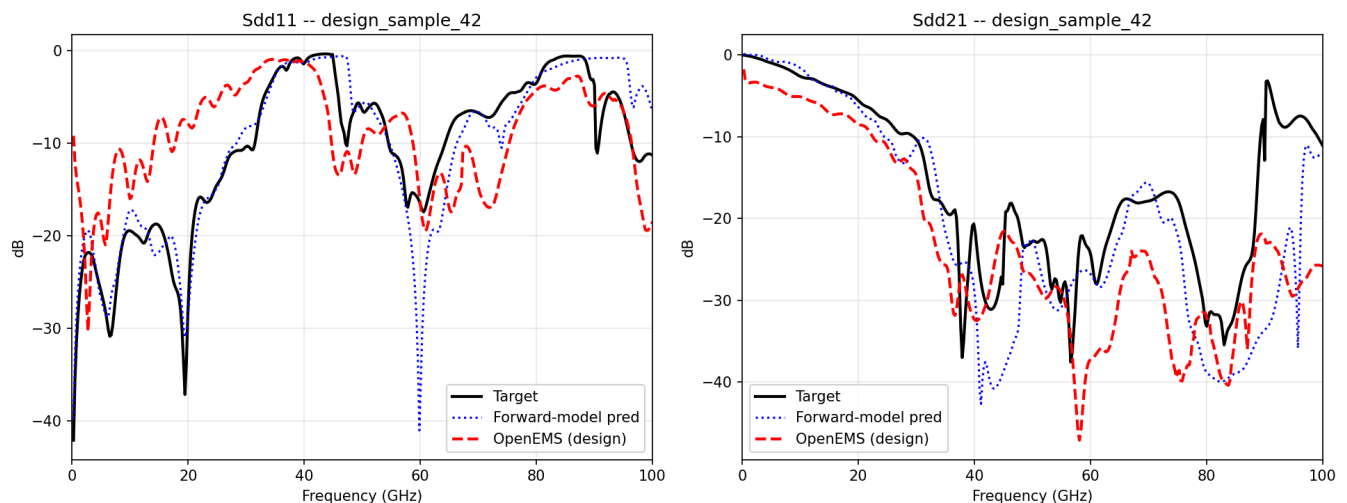
n_ports = 16

geometry looks physical (via < antipad < pitch/2, eps_r>1).

[--check-only] stopping before simulation. If the geometry looks good, re-run without --check-only.

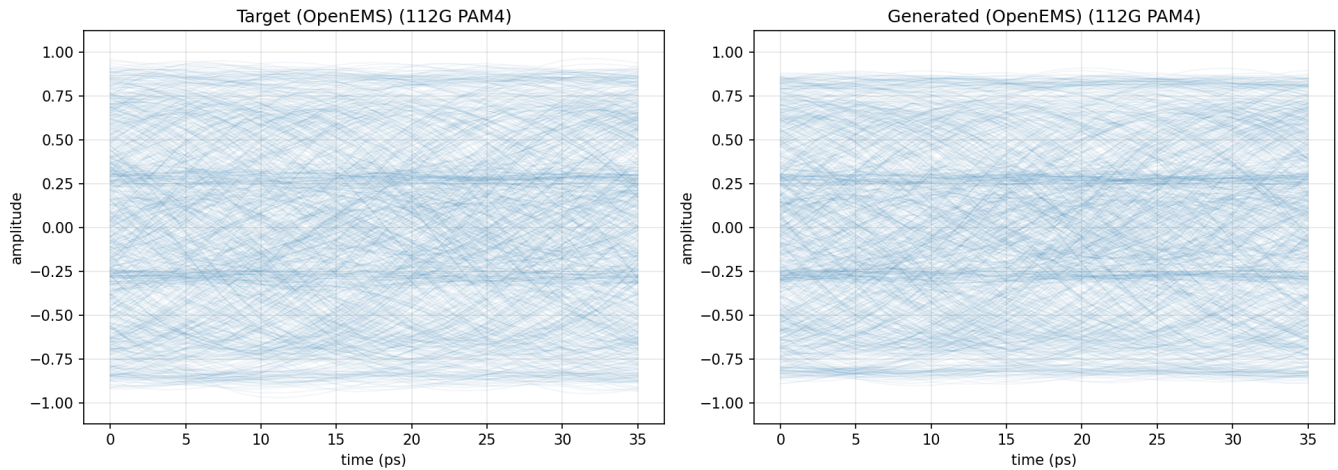


The sample 42 initially had a better fit when checked with the baseline model - and the openEMS validation gave comparatively better results as well. But still the need for latent TTO was clearly evident from the comparisons.



for sample 42 - which had less exploitation but still visible

112G PAM4 eye: Target (OpenEMS) vs Generated (OpenEMS) (total-opening diff 11.7%)



comparing eye diagram for target and generated for sample 0 - sim-0017

=====

Stage 08: 112G PAM4 eye (56 GBaud, UI = 17.857 ps, PRBS13Q, un-equalised)

=====

[Target (OpenEMS)] rail centres: [-0.6436, -0.214, 0.2184, 0.654]

[Target (OpenEMS)] eye heights (3 eyes): [0.1299, 0.1351, 0.1375]

[Generated (OpenEMS)] rail centres: [-0.6163, -0.1797, 0.2017, 0.6346]

[Generated (OpenEMS)] eye heights: [0.1369, 0.0842, 0.1344]

--- EYE AGREEMENT (target vs OpenEMS) ---

total eye opening [Target (OpenEMS)] : 0.4025

total eye opening [Generated (OpenEMS)]: 0.3555

total-opening difference : 11.7 %

per-sub-eye |height diff| : 0.0070, 0.0509, 0.0031

VERDICT: eyes differ by 11.7%

Latent Space TTO & Curriculum TTO combined (Final version for the Inverse model)

Inspired from Pascal Notin, José Miguel Hernández-Lobato, Yarin Gal, "Improving black-box optimization in VAE latent space using decoder uncertainty",

<https://doi.org/10.48550/arXiv.2107.00096>) for Latent Space TTO and (Liu, Z.; Shan, G.; Chen, Z.; Yang, Y. Physics-Guided Neural Surrogate Model with Particle Swarm-Based Multi-Objective Optimization for Quasi-Coaxial TSV Interconnect Design. *Micromachines* **2025**, *16*, 1134.

<https://doi.org/10.3390/mi16101134>) with passivity for Curriculum TTO.

in the baseline script the TTO descended directly on the geometry vector **x** and it drove $\tan \delta$ to 0.10, 0.33, 0.63 on samples 0/100/250, against a physical range of ~0.01–0.03 by cheating

the loss function to fit the curve. So I tried to optimize the latent space **z** instead of the geometry vector **x**. Now the Adam holds the latent space z (16-dimensions) adn the loss function now focus on the z score with no distortion and gradient exploitation.

When ran with the Latent TTO there was a dramatic improvement in the results and the geometry generated by the inverse model followed physical constraints and also generated manufacturable design metrics.

```
=====
===
Stage 07: design_sample_0_latent (template sim_pkg_0017, pair 1, geometry='x_local_norm')
=====
=====
```

--- geometry sanity check [design_sample_0_latent] ---
via_radius = 2.772 mil
antipad_rad = 18.416 mil
pitch = 53.026 mil
TDIEL/TMET = 503.1 total, eps_r=3.207, tan_d=0.0176
n_ports = 16
geometry looks physical (via < antipad < pitch/2, eps_r>1).

[--check-only] stopping before simulation. If the geometry looks good, re-run without --check-only.



Comparison table between values generated by geometry TTO model vs Latent Space TTO model. This is done for sim_pkg_0017

Geometry	Baseline (Geometry TTO)	Latent Space TTO
via_radius	2.403 mil	2.772 mil
antipad_rad	23.173 mil	18.416 mil
pitch	66.312 mil	53.026 mil
TDIEL/TMET	1296.2 total, eps_r=3.383, tan_d=0.1039	503.1 total, eps_r=3.207, tan_d=0.0176
n_ports	16	16

when comparing the data the Latent Space TTO gave a realistic design options with loss tangent $\tan_d=0.0176$.